# Bijux Pollenomics: scientific evidence

## Sources, method correspondence, predictions and model use

Bijux Pollenomics

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Source-bound research briefing

# Evidence ladder

1. Archive/source custody and taxon-count mapping.
2. Author-equation and numerical implementation parity.
3. Observed-pollen held-out prediction.
4. LRA reconstruction and forward-pollen prediction.
5. Independent measured-vegetation validation: no admitted pair yet.

Each rung answers a different question; a result at one rung does not imply the next.

Evidence: `reports/README.md`; `reports/needed/README.md`

# Four-country source and prediction accounting

| Target | Sites | Samples | Assigned/native NISP | Scored/attempted bins |
|---|---|---|---|---|
| Denmark | 1 | 91 | 122,998/129,768 | 24/24 |
| Finland | 28 | 1,797 | 1,229,019/1,260,281 | 435/435 |
| Norway | 52 | 3,544 | 2,018,295/2,286,020 | 1,084/1,178 |
| Sweden | 3 | 228 | 221,800/240,966 | 51/54 |

Closed 31-target author pollen composition. Excluded/unmapped mass remains outside the score. Zero Nordic samples are admitted to LRA by this study.

Evidence: `reports/science/denmark-finland-norway-author-pollen.md; reports/science/sweden-author-pollen.md`

# Source membership is distinct from a scored prediction

- We joined all 12,388 Neotoma source sample IDs to the four-country observed-pollen study: 5,660 selected across 84 sites, 6,218 outside those sites, and 277 without an assigned source country.
- The remaining 233 samples share selected sites but were not selected. The source pollen screen refuses all 233; age and count-candidate reasons overlap for six.
- Six selected Swedish samples also fail the source age screen. Their negative-BP ages map to nominal bin $-1$, which the frozen Swedish predictor refuses at all three sites; none is scored.

Study membership, source screening and scored site-bins are separate decisions. This is prior-exposed observed-pollen evidence, not LRA or vegetation validation.

Evidence: `reports/science/neotoma-study-membership/README.md`; `reports/science/neotoma-study-membership/report.json`

# SEAD is four-country chronology context

| Country | Assigned sites | Sites with claims | Chronology claims |
|---|---|---|---|
| Sweden | 1,925 | 741 | 22,643 |
| Denmark | 59 | 56 | 1,939 |
| Norway | 45 | 27 | 468 |
| Finland | 40 | 29 | 59 |

The 25,109 rows are chronology claims at 853 distinct sites, not pollen observations or admitted ecological events. A further 103 country decisions require review and 23 remain unassigned. All claim propagation is refused.

Evidence: `reports/science/source-cohorts.md`; `docs/report/source-relations/sead/chronology-country-accounting/report.json`

# Three further Swedish lake cores

- Metadata-only selection added Gilltjarnen, Lake Avegol and Flinkasjon outside the original Swedish targets.
- We assigned 231,115 of 238,797 native pollen grains to the 31 author targets; excluded and unmapped counts remain visible.
- All 63 site-bins received three fixed predictions, yielding 189 matched scores with no training-support refusal.
- Equal-site pollen MAE: IDW 0.017334, nearest site 0.017932, training mean 0.020682. Nearest site wins at Lake Avegol.

Three prior-exposed lake groups; direct observed pollen only, with no Nordic LRA or independent vegetation score.

Evidence: `reports/science/sweden-lake-transfer.md`; `docs/report/regions/nordic/swedish-lake-transfer/site-scores.csv`

# Swedish training scope on shared pollen targets

- Hold out each Swedish site; train on the other two Swedish sites or those same two plus 81 DK/FI/NO sites.
- On 39 shared site-bins, inverse-distance-squared equal-site pollen MAE is 0.032846 versus 0.023226, favoring four-country training.
- Sweden-only refuses 15 of 54 candidate bins; four-country training predicts all 54.

Three previously exposed sites; most mean gain comes from Abborrtjarnen. No LRA or independent vegetation accuracy follows.

Evidence: `reports/science/sweden-training-scope.md`; `docs/report/regions/nordic/swedish-training-scope/paired-scores.csv`

# Swedish source support changes the pollen score

- We froze 1,419 method predictions across 473 supported queries; 13 of 486 site-bin/policy queries refused.
- Adjacent nominal 500-year source bins and a nearest-five cap changed IDW error slightly and differently by site.
- Excluding source centers within 100 km raised matched equal-site IDW MAE from 0.023151 to 0.029171 on 52 bins.

Three exposed Swedish sites; site centers, no age posterior, LRA reconstruction or measured vegetation.

Evidence: `reports/science/sweden-spatiotemporal-sensitivity.md`; `docs/report/regions/nordic/swedish-support-sensitivity/paired-scores.csv`

# LegacyAge links two Nordic cores to alternative ages

- LegacyAge S5/S6 join 107/107 ABBO and 71/72 DALMUT depths; the unmatched 4.5 cm DALMUT row stays refused.
- All 178 matched Neotoma point ages lie within the reported 95% ranges; the age models differ.
- ABBO strict support: eight samples, four bins; site-excluded IDW MAE 0.060058 to 0.059762 on 12 paired targets.
- DALMUT strict support: two samples, six frozen 31-target Norway-excluded predictions; 20/22 bins unsupported.

Sparse prior-exposed development evidence. No historical LandClim age, LRA input or independent skill claim follows.

Evidence: `docs/report/source-relations/neotoma/legacyage-chronology/report.json`;

`docs/report/pollen-reconstruction/age-interval-sensitivity/comparison.json`;

`docs/report/regions/nordic/dalmutladdo-author-target-age-interval/report.json`

# Solsø: alternative ages restrict Danish pollen scoring

- Primary LegacyAge S5 joins all 92 depths across two Solsø cores; one Neotoma calibrated age stays refused.
- Strict interval containment retains 15/91 comparable samples in eight of 24 bins; both cores contribute.
- The 32 frozen Denmark-excluded predictions yield nearest-site eight-bin MAE 0.032310, versus 0.032104 on complete parent-bin counts.
- Separately, 24 frozen 31-target predictions give nearest-site MAE 0.025673 on author-assigned counts; the axes and denominators differ.

One prior-exposed lake; 16-label and 31-target pollen scores use different denominators. No age reassignment, historical LandClim selection, LRA or vegetation validation follows.

Evidence: `docs/report/countries/denmark/solsoe-source-lineage/legacyage-chronology/report.json`;

`docs/report/countries/denmark/solsoe-source-lineage/legacyage-interval-sensitivity/report.json`;

`docs/report/countries/denmark/solsoe-source-lineage/author-target-age-interval/report.json`

# Swedish chronology-block pollen transfer

- Hold out each target site; predict 54 target bins using training pollen exactly 1,000 nominal BP older, excluding the target and adjacent source bins.
- All 54 bins yielded three frozen predictions and matched same-bin scores; zero queries refused.
- Inverse-distance-squared equal-site pollen MAE changed from 0.023647 to 0.024023 on the same targets; nearest-site and training-mean changes were small and in the opposite direction.

Prior-exposed three-site development result; assigned age bins do not account for age-model uncertainty. No Nordic LRA or measured vegetation score follows.

Evidence: `reports/science/sweden-chronology-block.md`; `docs/report/regions/nordic/swedish-chronology-block/paired-scores.csv`

# A newly scored Swedish lake on a different pollen axis

- Lake Sämbosjön was absent from earlier targets; all 22 candidate bins received three fixed predictions and scores.
- Inverse-distance-squared equal-bin pollen MAE was 0.022991 on 16 exact source labels, covering 224,291 of 393,822 native pollen counts.
- Aggregate count support was screened before prediction; aggregate scores were viewed before a later protocol annotation correction. Prediction bytes did not change.
- Source age bounds place 58 of 111 samples across a nominal 500-year bin boundary; all 22 scored bins contain a crossing sample.

One-site descriptive development result. The 16-label error is not comparable to the 31-author-target error; no independent confirmation, LRA or measured vegetation score follows.

Evidence: `reports/science/sambosjon-direct-pollen.md`; `docs/report/regions/nordic/sambosjon-direct-pollen/report.json`; `docs/report/regions/nordic/sambosjon-direct-pollen/age-bounds/report.json`

# A second Swedish lake: fixed methods differ by metric

- Lilla Gloppsjön site/core was excluded from 84 training sites; 23 bins received 69 predictions before target scoring.
- All 69 scores completed on 16 exact pollen labels: 55,638 of 127,027 native NISP counts lie on the closed response.
- Equal-bin MAE: training mean 0.016272, primary inverse-distance-squared 0.017295, nearest site 0.020000.
- Hellinger: nearest site 0.228724, inverse-distance-squared 0.237576, training mean 0.256392. No method was selected using this target.

One-site descriptive direct-pollen development; the earlier source-validation screen parsed the target partition. Two age-refused samples remain outside scores; no independent LRA or vegetation claim.

Evidence: `reports/science/lilla-gloppsjon-direct-pollen.md`; `docs/report/regions/nordic/lilla-gloppsjon-direct-pollen/scored/report.json`

# What reproduces numerically

- Author raw-to-prepared replay: 17 example files, 871 raw sample rows, 15,028 matching prepared cells.
- WHH05: 301 raw grains map to 297 prepared; 31 paired means and SEs agree at 1500 BP.
- Reexecuted author R/native calculations on two grids (9+8 sites); 14,074 supported mean and SE values agree within declared tolerances.
- Archive TW2 differs: total variation 0.092/0.111; two taxa explain 73%/94% of each grid's L1 gap. Historical inputs remain unclosed.
- Archived ISTS and TBS mean arithmetic closes over all 10,500 grid-window rows, but biological classification and SE transformation remain open.

Evidence: `reports/science/publication-correspondence.md`; `reports/science/publication-correspondence/published-window-provenance.md`; `reports/science/publication-correspondence/author-example-reexecution.md`

# LOVE 2024: paper inputs and unavailable output

- All 30 published fall-speed, PPE and PPE-SD values match the later DISQOVER example by row position; nine of ten taxon labels match exactly.
- The paper's eight simulated ponds, landscape extraction, ring geometry, pollen inputs and 100 random runs are not present in the retained example.
- Figure 2 RMSE and its 100–500 m tendency therefore remain a paper result, not a reproduced Nordic model setting.

Input correspondence is established; the eight-site numerical simulation is blocked on its original inputs.

Evidence: `reports/science/publication-correspondence.md`; `docs/report/method-certification/love-paper-correspondence/README.md`

# LOVE optimizers: shared objectives, different searches

- We gave installed RcppDE and Python the same three declared pollen, deposition, background, PPE and bound profiles.
- Both attained objective gaps below $10^{-4}$ from independent optima; cross-evaluating selected candidates differed by less than $10^{-10}$.
- Python reported success after 29–40 iterations and 600–1,230 evaluations. R used its full 400-iteration cap in all three cases.
- One objective has different, equally good vegetation vectors; vector equality would misstate solution quality.

Bounded objective and arithmetic checks; no identical random path, general global guarantee or ecological accuracy.

Evidence: `reports/science/method-correspondence.md`; `docs/report/method-certification/optimizer-solution-quality/summary.csv`

# One fixed LOVE example in both source engines

- R 4.6.0 / DISQOVER 0.9.14 and current Python read one shared `LOVEexample` input set. Hexadecimal readback verified every numeric input cell.
- Each engine evaluated 32 fixed candidates on eight active ring/repeat calls. All eight selected candidate indices agreed.
- Mean cover and pollen matched exactly on 40 finite cells each. Ring-factor and objective differences were at most $5.11 \times 10^{-15}$ on 360 and eight finite cells, respectively.

This checks fixed-candidate source arithmetic, not search convergence, stochastic draws, timing or ecological accuracy.

Evidence: `docs/report/numerical-performance/love-source-comparison/report.json`

# English LOVE: declared model profile

- 15 source sites, 1250–1750 BP, 30 targets; published PPE and fall speeds held fixed for retained taxa.
- Regional REVEALS plus ring-taxon-preserving LOVE; closed, nonnegative local cover.
- Uniform-start SLSQP: 300 iterations, 16,000 evaluations, objective tolerance $10^{-8}$.
- 192/195 training fits complete; three failures also refused under a bounded 600-iteration check.

These are source-specific development choices, not universally optimal LRA parameters or the original R optimizer.

Evidence: `reports/science/english-lra-pollen-prediction.md`

# English LOVE routes have different support

| Route | Source axis and attempts | Retained result |
|---|---|---|
| Analytical | 31 taxa; 15 folds | 15/15 invalid compositions; all 54 local ring rows include negative cover. |
| Differential evolution | 30 taxa; one pair, two requests | Both requests refused convergence; no held-out score. |
| Constrained SLSQP | 30 taxa; 15 folds | 14/15 folds predicted; BECKHAM had three training-fit refusals. |

Excluding 159 Filipendula counts changed the taxon response. These development runs are not a paired solver ranking or a vegetation-accuracy comparison.

Evidence: `reports/science/english-lra-pollen-prediction.md`

# Matched pollen comparison: result and limitation

| | MAE | | Hellinger | |
|---|---|---|---|---|
| | Direct | LRA | Direct | LRA |
| Training mean | .020496 | .026273 | .282082 | .360671 |
| Nearest | .019552 | .021132 | .255767 | .286001 |
| Inverse distance squared | .018579 | .021088 | .243673 | .285415 |

14 paired English sites, 30 taxa; all three direct means remain lower after any one site is omitted. Prior exposure and unequal training information limit interpretation.

Evidence: `docs/report/pollen-reconstruction/author-lra-constrained/direct-pollen/paired-hellinger.json`

# English basin radius: forward sensitivity only

- We replayed 42 frozen LRA vegetation estimates at 80%, 100% and 120% of each assumed target-lake radius.
- The actual forward consumer reproduced all 42 original pollen predictions exactly before target scoring.
- Inverse-distance-squared site-macro pollen MAE: .021032, .021088 and .021457 across those radii.
- On matched targets, direct pollen retains lower mean MAE and Hellinger for every method and radius factor.

14 prior-exposed English sites. Vegetation, regional cover and PPE were fixed; this does not calibrate basin geometry or admit Nordic LRA inputs.

Evidence: `docs/report/pollen-reconstruction/author-lra-constrained/radius-sensitivity/paired/report.json`; `reports/science/english-lra-pollen-prediction.md`

# Model selection follows the target and support

- Fixed spatial baselines permit clear paired interpretation on observed pollen.
- A bounded RBF candidate improved one prior-exposed author comparison on common support but reduced coverage.
- Country-excluded tuning did not establish superiority over inverse-distance-squared in the Nordic development holdouts.
- A two-case synthetic PPE/transport winner reflects a transport-law mismatch and is not a real-core parameter recommendation.
- For LRA, input identity, chronology, basin geometry, PPE and denominator must be qualified before any parameter search is meaningful.

Evidence: `reports/science/model-use.md`; `reports/science/synthetic-model-selection/README.md`; `reports/needed/README.md`

# HYDE and vegetation: retain the boundary

- Four-country HYDE source cropland grid sums match 84 country/year/scenario totals within $8.51 \times 10^{-11}$ km$^2$.
- This checks modeled-source internal accounting, not pollen-model benefit.
- The retained release has conflicting rights and version declarations; formal CRS and historical calendar support remain unresolved.
- No independent vegetation measurement pair has been admitted for prediction accuracy.

Next experiments require source semantics and a matched ecological reference, with outcomes protected from model selection.

Evidence: `reports/science/hyde-cropland-source-account.md`; `reports/needed/README.md`

# Why no vegetation-accuracy score exists yet

- EMPD2–NFI screen: 16 nearby candidates, zero measured pairs admitted for our benchmark.
- Färskesjön: the measured 7 m forest plot is 427 m from the lake pollen point; five volume classes do not supply 31-taxon landscape cover.
- Three Småland bogs: historical-map coverage and pollen anchors are published, but full numerical vegetation fractions are absent.
- A published southern-Swedish figure favors its authors' REVEALS estimates on nine rounded printed labels; it does not score our predictions or recover the full 26-taxon study.
- LandClim grid 779 names Bussjösjön and Bökesjön as input sites; its modeled cover is not an independent measured target for those pollen series.

A physical-site, time, taxon and landscape-support match is still required before ecological validation.

Evidence: `reports/science/vegetation-reference.md`; `docs/report/countries/sweden/pollen-model-lineage/report.json`; `docs/report/validation/empd2-nfi-pairing/report.json`

# Open source decisions

- Abborrtjärnen: LandClim-selected age model, historical basin radius, eligible denominator, PPE use and rights.
- Bjäresjösjön: modern lake identity and nearby SEAD display intervals close; pollen-core identity and ecological event join remain refused.
- Solsø: 79 LandClim II samples versus 92 Neotoma source samples; the published age span retains 90, and no historical basin series is bound to the selected rows.
- Dalmutladdo (Norway): 72 source samples and 11 radiocarbon controls align; published 7.2 ha versus LandClim 8 ha, with no dated basin series. Zero LRA admission.
- Laihalampi (Finland): one LAIH core and 150 depths align; modern area claims agree conditionally, but selected ages and dated basin remain unqualified.
- Historical publication: exact source revisions and authenticated PFT/uncertainty transformation.
- Independent confirmation: new qualified cohort frozen before target inspection.

Evidence: `reports/needed/README.md`; `reports/science/solsoe-lra-input.md`; `reports/science/dalmutladdo-lra-input/README.md`; `reports/science/laihalampi-lra-input/README.md`